



The Open
University

M303

Module Examination 2025
Further Pure Mathematics

Tuesday 16 September 2025

There are **two sections** in this examination.

In **Section 1** you should **submit answers to all 6 questions**. Each question is worth 11% of the total mark. You are advised to spend about two hours on this section.

In **Section 2** you should **submit answers to 2 out of the 3 questions**. Each question is worth 17% of the total mark.

Do not submit more than the required number of answers in Section 2. If you do, only your first 2 answers will be marked.

Include all your working, as marks may be awarded for it.

Use the methods and results taught in M303 rather than any alternative methods or results not covered in M303.

Start your answer to each question on a new page, clearly indicating the number of the question.

Crossed out work will not be marked.

When sketching sets, take care to indicate which points are included: use dashed lines to indicate boundaries that are not included and solid lines to indicate boundaries that are. Use small open circles $\circ$ to indicate any 'corner' points on a boundary that are not included and small closed circles $\bullet$ to indicate any 'corner' points that are included.

Follow the instructions in the online timed examination for how to submit your work.

Further information about completing and submitting your examination work is in the *Instructions and guidance for your remote examination* document on the module website.

Submit your exam using the iCMA system (iCME81). Make sure that the name of the PDF file you submit includes your PI and the module code e.g. X1234567M303.

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QP ID: M3032509I4PV1

Section 1

You should **submit answers to all questions**.
Each question is worth 11%.

Question 1 (11 marks)

- (a) Let $a_1 = -2, a_2 = 3$ and $a_3 = 5$.
Let $a_n = a_{n-3} + 2a_{n-2} + 2a_{n-1}$ for all $n \geq 4$.
Prove using the Second Principle of Mathematical Induction that
for all $n \geq 1$, we have $a_n \leq \frac{1}{2} \times 3^n$. [6]
- (b) Prove that for all $m \in \mathbb{Z}$, the numbers $3m + 1$ and $11m + 4$ are
coprime. [2]
- (c) Prove that $n^7 + 7n^6 - 21n^2 + 13n$ is divisible by 14 for all n . [3]

Question 2 (11 marks)

- (a) Let $G = \mathbb{Z}_{10} \times \mathbb{Z}_5$.
(i) Write down the order of G . [1]
(ii) Write down the order of the element $(1, 0)$ of G . [1]
(iii) Write down an element of G of order 2. [1]
- (b) Let M be the group defined as follows. The underlying set of M is
the set of all 2×2 matrices with coefficients in $\mathbb{Z}_{10}$, that is

$$M = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \right\}.$$

The operation is $\oplus$ where $\oplus$ is given by:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \oplus \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} a + a' & b + b' \\ c + c' & d + d' \end{pmatrix}$$

where $+$ is addition modulo 10.

You may assume that this is a group. You do not have to prove
this.

- (i) Write down the identity element of M and the inverse of
 $\begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$. [1]
- (ii) Let H be defined as

$$H = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : a, b \in \mathbb{Z}_{10} \right\}.$$

Show that H is a subgroup of G . [2]

- (c) Consider the following two groups, G and H , both of which have
order $3861 = 3^3 \times 11 \times 13$:

$$G \cong \mathbb{Z}_{33} \times \mathbb{Z}_3 \times \mathbb{Z}_{39},$$

$$H \cong \mathbb{Z}_{99} \times \mathbb{Z}_{39}.$$

- (i) Write G as a direct product of cyclic groups of prime power order.
- (ii) Write H as a direct product of cyclic groups of prime power order.
- (iii) Write down the order of the largest cyclic subgroup of G .
- (iv) Is H cyclic? Justify your answer.
- (v) Write down, as a direct product of cyclic groups of prime power order, an abelian group of order 3861 that is not isomorphic to either G or H . [5]

Question 3 (11 marks)

- (a) Given $6171 = 3 \times 11^2 \times 17$, determine the following (showing your workings clearly).
 - (i) the divisors of 6171
 - (ii) $\tau(6171)$
 - (iii) $\sigma(6171)$
 - (iv) $\phi(6171)$. [4]
- (b) Factorise $f(x) = x^4 + 10x^3 - 12x^2 + 40x + 22$ into a product of irreducible polynomials over $\mathbb{Z}_3$. [4]
- (c) Determine all primes p such that $\sigma(p^3) - \phi(p^3)$ is prime. [3]

Question 4 (11 marks)

Let

$$A = \{(a, b) \in \mathbb{R}^2 : a^2 + (b - 2)^2 \leq 4\},$$

$$B = \{(a, b) \in \mathbb{R}^2 : b - |a| > 0\} \text{ and}$$

$$C = A \cap B.$$

- (a) Sketch on separate diagrams or describe in words each of A , B and C taking care to indicate which points lie in the sets. [3]
- (b) Determine whether or not $(0, 0)$ is a closure point of C for the Euclidean metric $d^{(2)}$. [2]
- (c) Write down in set notation the closure, interior and boundary of C for the Euclidean metric $d^{(2)}$. [4]
- (d) Determine whether or not $(0, 0)$ is a boundary point of C for the ‘mixed’ metric d on $\mathbb{R}^2$ given by

$$d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| + d_0(x_2, y_2),$$

where d_0 is the discrete metric on $\mathbb{R}$. [2]

Question 5 (11 marks)

Let $\alpha = \sqrt{11} - \sqrt{2}$.

- (a) Express α^3 in the form $a\sqrt{11} + b\sqrt{2}$ for suitable $a, b \in \mathbb{Q}$.
- (b) With the help of (a) or otherwise, show that $\sqrt{11}, \sqrt{2} \in \mathbb{Q}(\alpha)$.
- (c) Hence show that $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{11}, \sqrt{2})$. [1]
- (d) By considering α^2 find the minimal polynomial for α over $\mathbb{Q}$. You may assume that $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. [3]
- (e) Prove that $\mathbb{Q}[x]/\langle 2x^5 + 6x^4 - 3x^3 + 27x - 6 \rangle$ is a principal ideal domain. [3]

Question 6 (11 marks)

- (a) Let $A = \{(a, b) \in \mathbb{R}^2 : |a - b| \leq 3\}$. Determine whether A is a $d^{(2)}$ -compact subset of $\mathbb{R}^2$.
- (b) Let $X = \mathbb{R}^2 - \{(a, b) \in \mathbb{R}^2 : a = 0\}$ and let d be the metric on X given by

$$d(\mathbf{x}, \mathbf{y}) = 2|x_1 - y_1| + 3\sqrt{|x_2^2 - y_2^2|},$$

where $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ are points in $\mathbb{R}^2$.

(You do not have to show that d is a metric.)

- (i) Show that

$$d((1/n, 0), (a, b)) \geq 2(|a| - 1/n) \text{ for } (a, b) \in X \text{ and } n \in \mathbb{N}.$$

- (ii) Hence, or otherwise, show that $((1/n, 0))_{n=1}^\infty$ is not a d -convergent sequence in X .
- (iii) Hence, or otherwise, show that X is not d -complete.
- (iv) Deduce that X is not d -sequentially compact.
- (c) Let

$$A = \{(t^2, t^3) \in \mathbb{R}^2 : 0 \leq t \leq 1\}$$

and define $f: [0, 1] \rightarrow \mathbb{R}^2$ by $f(t) = (t^2, t^3)$.

- (i) Show that f is $(d_{[0,1]}^{(1)}, d^{(2)})$ -Lipschitz.
- (ii) Hence, or otherwise, show that A is $d^{(2)}$ -connected.

Section 2

You should **submit answers to two questions**. If you submit more, then only the scores from the first two answers submitted will count. **Each question is worth 17%.**

Question 7 (17 marks)

- (a) Let G be a group of order $30752 = 2^5 \times 31^2$.
- (i) Write down the orders of the Sylow 2-subgroups and the Sylow 31-subgroups. [2]
 - (ii) Determine the possible numbers of Sylow 2-subgroups and Sylow 31-subgroups being careful to specify which results you are using in your reasoning. [5]
- (b) Consider the dihedral group of order 62 given by:
- $$D_{31} = \langle r, s \mid r^{31} = s^2 = e, sr = r^{30}s \rangle.$$
- (i) Write the rsr^2 of D_{31} in standard form. [2]
 - (ii) Let H be the group of order 30752 given by:
 $H \cong D_{31} \times D_{31} \times \mathbb{Z}_n$ that is the direct product of the Dihedral group D_{31} with itself and with a cyclic group, $\mathbb{Z}_n$.
Show that $n = 8$. [1]
 - (iii) Are the Sylow 2-subgroups of H abelian? Justify your answer. [2]
- (c) Let K be the group of order 30752 given by: $K \cong D_{31} \times D_{31} \times D_4$ that is the direct product of the Dihedral group D_{31} with itself and with the dihedral group D_4 .
- (i) Write down a group isomorphic to a Sylow 2-subgroup of K . [1]
 - (ii) Are the Sylow 2-subgroups of K abelian? Justify your answer. [1]
- (d) Let G be a group with identity element e . Define the set S by $S = \{g \in G : g^5 = e\}$. Prove that S cannot have even order. [3]

Question 8 (17 marks)

- (a) (i) Given that $731021 = 857 \times 853$ calculate $(731021/859)$. You may assume that 853, 857 and 859 are prime and you should clearly reference any results used. [3]
- (ii) Let p be a prime that is equal to $2n + 1$ for some n that is congruent to 5 (mod 8). Prove that 2 is not a quadratic residue of p . [2]
- (b) Let $R = \mathbb{Z}(\sqrt{5})$. We define a norm N on R by $N(a + b\sqrt{5}) = |a^2 - 5b^2|$. You may assume that N is multiplicative and that every element of R can be written uniquely as $a + b\sqrt{5}$ with $a, b \in \mathbb{Z}$. You may assume that $a + b\sqrt{5}$ is a unit of R if and only if $N(a + b\sqrt{5}) = 1$.
- (i) Calculate $N(9 + 4\sqrt{5})$ and $(9 + 4\sqrt{5})^2$. [1]
- (ii) Explain why $N((9 + 4\sqrt{5})^2) = 1$. [1]
- (iii) Let $a, b, c, d > 0$ be positive integers. Let $(a + b\sqrt{5})(c + d\sqrt{5}) = x + y\sqrt{5}$. Explain why $x > a$. [1]
- (iv) Deduce that R has infinitely many units. [2]
- (v) Prove that $\langle 2, 1 + \sqrt{5} \rangle \neq R$. [4]
- (c) Let D be an integral domain. Let a be a non-zero element of D such that adding a to itself 5 times gives zero (in other words $a + a + a + a + a = 0$). Prove that for all $b \in D$ adding b to itself five times also gives zero. [3]

Question 9 (17 marks)

- (a) Let
- $$A = \{f \in C[0, 1] : |f(x) - f(y)| \leq |x - y|^{1/3} \text{ for each } x, y \in [0, 1]\}$$
- and
- $$B = \{f \in C[0, 1] : |f(x)| \leq 2 \text{ for each } x \in [0, 1]\}.$$
- (i) Determine if A is a pointwise bounded set. [1]
- (ii) Show that A is a $(d_{[0,1]}^{(1)}, d^{(1)})$ -equicontinuous set of functions. [2]
- (iii) Show that A is a $d_{\max}$ -closed set of functions. [2]
- (iv) Show that $A \cap B$ is $d_{\max}$ -sequentially compact. [2]

- (b) Let $d: [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ be a distance function given by

$$d(x, y) = |x^4 - y^4| + d_0(x^3, y^3), \quad \text{for } x, y \in \mathbb{R},$$

where d_0 is the discrete metric on $\mathbb{R}$.

Show that d is a metric on $\mathbb{R}$.

[6]

- (c) Suppose that $T: [0, 1] \rightarrow [0, 1]$ is a contraction mapping for the metric d given in part (b).

Show that if T has contraction ratio $\lambda < 1/2$, then T is a constant function.

[4]

[END OF QUESTION PAPER]